Guide to Changing All System Audio Files to Arabic

1. Introduction

In this guide, you will learn how to change all system audio files to Arabic using the **MultilingualResourceTool** and integrating it with a Text-to-Speech service.

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2. Steps to Follow

Step 1: Download MultilingualResourceTool

- Access the following link to download the tool: MultilingualResourceTool on GitHub.

Clone or download the source code:
git clone https://github.com/marklogg/ubt_alphamini_multilingualtool.git

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Step 2: Install Python

- **MultilingualResourceTool** is written in Python, so you need to install Python first.
- Download Python from python.org and follow the installation instructions for your operating system.

Install the required libraries using the following command in the project directory:
pip install -r requirements.txt

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Step 3: Read the README File

- Open the `README.md` file in the project directory to understand how to use the tool and its specific requirements.

Step 4: Modify the Excel File to Add an Arabic Column

- Locate the language management Excel file in the project directory.
- Add a new column to the table and name it "Arabic".
- Translate all content into Arabic and input it into the new column.

Step 5: Find a Text-to-Speech Service Supporting Arabic

- Search for a Text-to-Speech service that supports Arabic, such as:
  - **Google Cloud Text-to-Speech**
  - **Amazon Polly**
  - **Microsoft Azure Cognitive Services**

Step 6: Modify the Code to Integrate the TTS API

Use the following code snippet to handle text-to-speech conversion:

```
 import pandas as pd
from google.cloud import texttospeech

def text_to_speech(file: str, text: str) -> bool:
    """
    Use Google Text-to-Speech API to convert text into an MP3 file.

    :param file: Name of the file to save
    :param text: Text to convert
    :return: True if successful, False otherwise
    """
    try:
        client = texttospeech.TextToSpeechClient()
        synthesis_input = texttospeech.SynthesisInput(text=text)

        voice = texttospeech.VoiceSelectionParams(
            language_code="ar-XA",
            ssml_gender=texttospeech.SsmlVoiceGender.NEUTRAL
        )

        audio_config = texttospeech.AudioConfig(
            audio_encoding=texttospeech.AudioEncoding.MP3
        )

        response = client.synthesize_speech(
            input=synthesis_input, voice=voice, audio_config=audio_config
        )

        with open(file, "wb") as out:
            out.write(response.audio_content)
            print(f"Audio content written to file {file}")
        return True
    except Exception as e:
        print(f"Failed to synthesize text: {e}")
        return False

def create_localtts_files(it):
    text = it["Arabic"]
    file_name = it["Audio File Name"]
    if text:
        if text_to_speech(file=file_name, text=text):
            print(f"\"{text}\" to speech-file: {file_name} success!")
        else:
            print(f"Failure! Please check the text-to-speech integration.")

def main():
```

```
    t = pd.read_excel("./excels/language_localTts.xlsx", header=1)
    print(t.info())
    t.apply(create_localtts_files, axis=1)

if __name__ == '__main__':
    main()
```

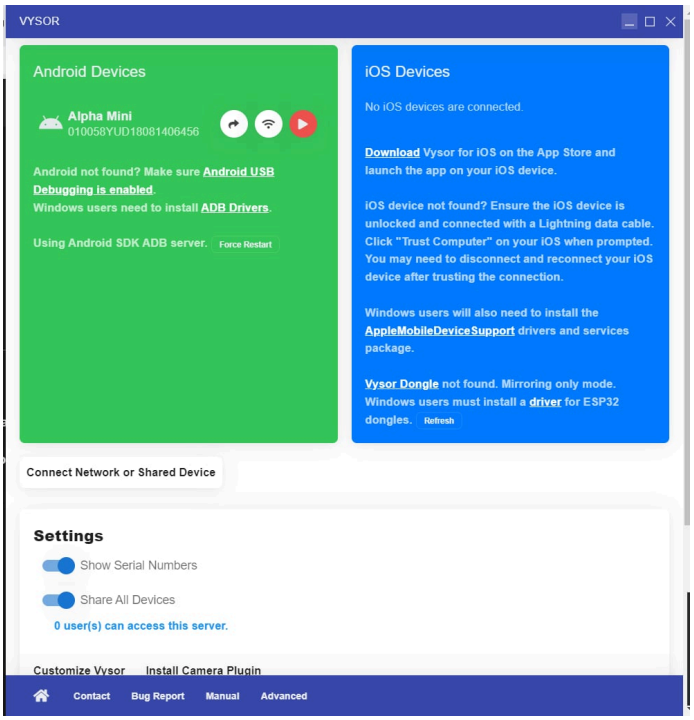
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Step 7: Generate MP3 Files

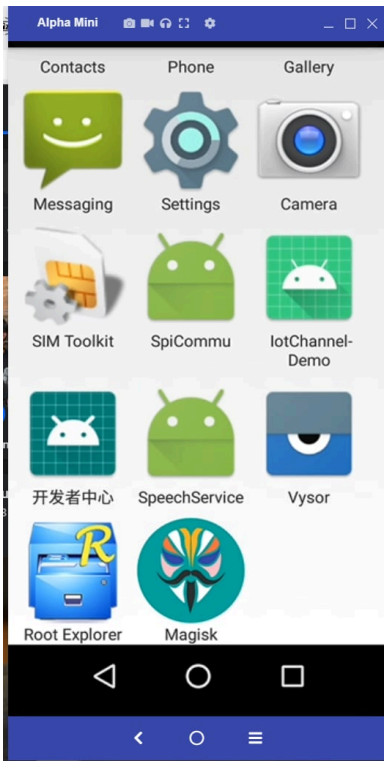
- Use the modified code to generate MP3 audio files from the Arabic content in the Excel file.
- Save the MP3 files in a format compatible with the robot's system.

Step 8: Transfer Audio Files to the Robot System

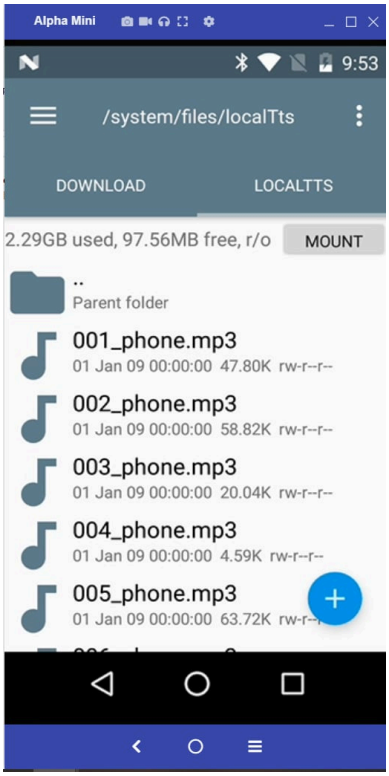
- Use **Vysor** to control the robot system. Download Vysor from vysor.io/download.
- Connect the robot to your computer using a USB cable. The robot must:
    - Be **unlocked**.
    - Have **USB Debugging** enabled.
- Open Vysor and remote into the robot system.



- On the robot, select the file transfer mode to allow your computer to access the robot's SDCard.
- Copy all MP3 files to the **Download** directory on the robot.



- On the robot, open the **Root Explorer** app (pre-installed).
    - Use the app to copy the MP3 files from the **Download** directory to the **/system/files/localTts** directory.
    - Overwrite the old system MP3 files.



- Complete the audio file replacement process.

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3. Conclusion

By following the above steps, you can successfully replace all system audio files with Arabic versions. Make sure to thoroughly test the audio files after conversion to ensure smooth operation on the robot's system.

Building the Speech Demo Framework to add Arabic

In the Development Guide of Speech Service for Alpha Mini, in step 3.3, if you wish to use the default wakeup module provided by the supplier, called "Hey Mini", for your speech demo framework, you need to contact UBTECH to request a license for this module. A sample license is included in the document. If you are granted the license, please place it in the `sdcard/weinalicence` directory.

If you are unable to obtain the license, you will be required to create your own wakeup module using a third-party solution.